ISWE406P - AGILE DEVELOPMENT PROCESS AND DEVEOPS

LAB FAT EXAM

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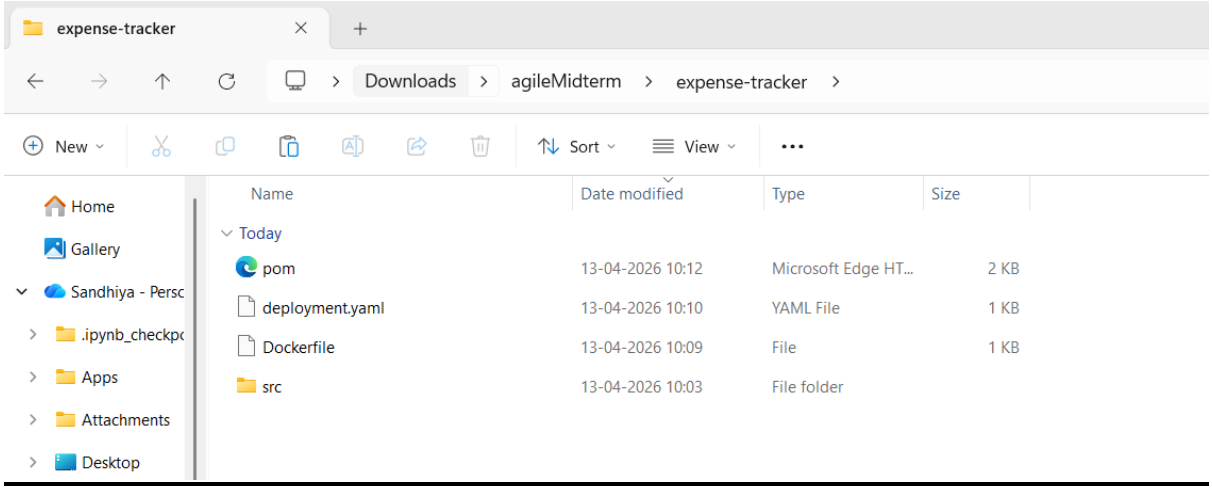
Expense Tracking System

Simple Java Application (in local machine)



Java Files

1. ExpenseManager.Java

```
package com.expense;
import java.util.*;
class Expense {
    String category;
    double amount;

    Expense(String category, double amount) {
        this.category = category;
        this.amount = amount;
    }
}
public class ExpenseManager {

    private List<Expense> expenses = new ArrayList<>();

    public void addExpense(String category, double amount) {
        expenses.add(new Expense(category, amount));
    }

    public double getTotalExpense() {
        double total = 0;
        for (Expense e : expenses) {
            total += e.amount;
        }
        return total;
    }
```

```
    public double getExpenseByCategory(String category) {
        double total = 0;
        for (Expense e : expenses) {
            if (e.category.equalsIgnoreCase(category)) {
                total += e.amount;
            }
        }
        return total;
    }
    public void clearExpenses() {
        expenses.clear();
    }
    public static void main(String[] args) {
        ExpenseManager manager = new ExpenseManager();
        manager.addExpense("Food", 200);
        manager.addExpense("Travel", 300);

        System.out.println("Total Expense: " + manager.getTotalExpense());
    }
}
```

ExpenseManagerTest.java

```
package com.expense;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;
```

```
public class ExpenseManagerTest {

    @Test
    void testTotalExpense() {
        ExpenseManager manager = new ExpenseManager();
        manager.addExpense("Food", 200);
        manager.addExpense("Travel", 300);

        assertEquals(500, manager.getTotalExpense());
    }

    @Test
    void testCategoryExpense() {
        ExpenseManager manager = new ExpenseManager();
        manager.addExpense("Food", 200);
        manager.addExpense("Food", 100);

        assertEquals(300, manager.getExpenseByCategory("Food"));
    }

    @Test
    void testClearExpenses() {
        ExpenseManager manager = new ExpenseManager();
        manager.addExpense("Food", 200);
        manager.clearExpenses();

        assertEquals(0, manager.getTotalExpense());
    }
}
```

<u>Maven File</u> - pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.expense</groupId>
    <artifactId>expense-tracker</artifactId>
    <version>1.0</version>
    <packaging>jar</packaging>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
        <junit.version>5.10.1</junit.version>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>${junit.version}</version>
            <scope>test</scope>
        </dependency>
    </dependencies>
```

```
<build>
    <plugins>

        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-compiler-plugin</artifactId>
            <version>3.11.0</version>
        </plugin>

        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-surefire-plugin</artifactId>
            <version>3.2.5</version>
            <configuration>
                <useModulePath>false</useModulePath>
            </configuration>
        </plugin>

        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-jar-plugin</artifactId>
            <version>3.4.1</version>
            <configuration>
                <archive>
                    <manifest>
                        <mainClass>com.cart.Cart</mainClass>
                    </manifest>
                </archive>
```

```
            </configuration>
        </plugin>

    </plugins>
  </build>
</project>
```

Kubernetes Deployment File deployment.yaml

```yaml
deployment.yaml
  spec:
    replicas: 2
    selector:
      matchLabels:
        app: expense-app
    template:
      metadata:
        labels:
          app: expense-app
      spec:
        containers:
        - name: expense-app
          image: sandhiya1701/expense-tracker:latest
          imagePullPolicy: Always
```

Dockerfile

```dockerfile
Dockerfile

FROM eclipse-temurin:17-jdk-alpine
WORKDIR /app
COPY target/*.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "app.jar"]
```

Build Locally using Maven (Test)

Inside project folder:

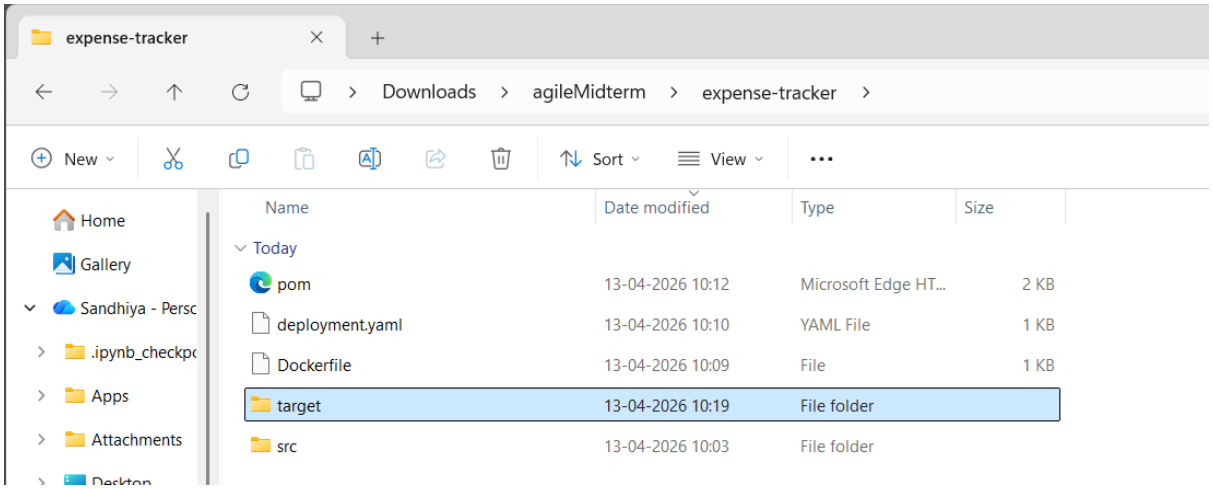
mvn clean test

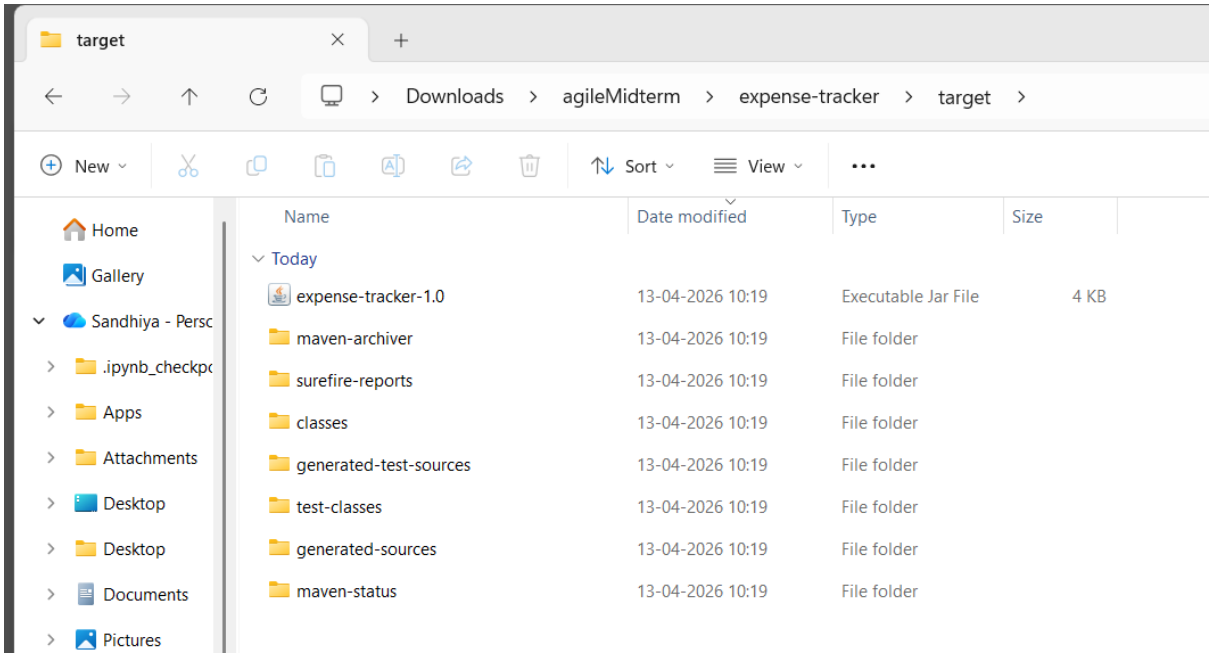
```
PS C:\Users\sandh\Downloads\agileMidterm\expense-tracker> mvn clean test
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.expense.ExpenseManagerTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.052 s
 in com.expense.ExpenseManagerTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.028 s
[INFO] Finished at: 2026-04-13T10:19:00+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\sandh\Downloads\agileMidterm\expense-tracker>
```

mvn clean package

```
PROBLEMS 13   OUTPUT   TERMINAL   ...                powershell

PS C:\Users\sandh\Downloads\agileMidterm\expense-tracker> mvn clean package
[INFO] Running com.expense.ExpenseManagerTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.053 s --
 in com.expense.ExpenseManagerTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ expense-tracker ---
[INFO] Building jar: C:\Users\sandh\Downloads\agileMidterm\expense-tracker\target
\expense-tracker-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.902 s
[INFO] Finished at: 2026-04-13T10:19:57+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\sandh\Downloads\agileMidterm\expense-tracker>
```

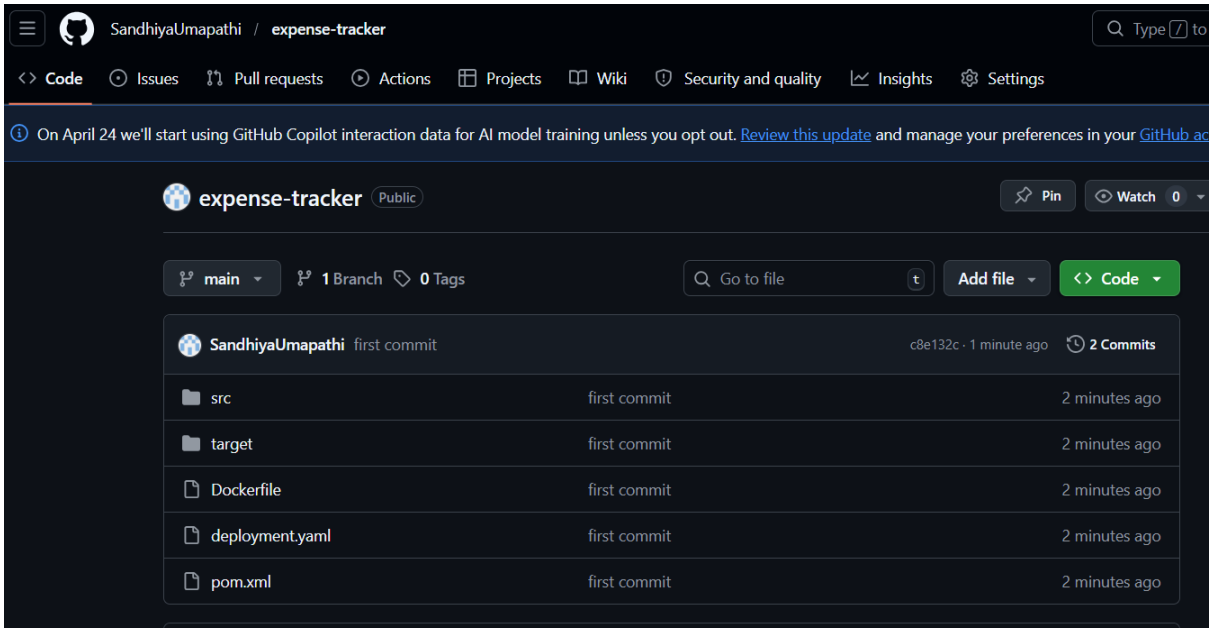
It generates target file





Push Code to GitHub

```
git init
git add .
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/SandhiyaUmapathi/expense-tracker.git
git push -u origin main
```



Jenkins Pipeline Job

```
pipeline {
    agent any

    tools {
        jdk 'JDK17'
        maven 'Maven3'
```

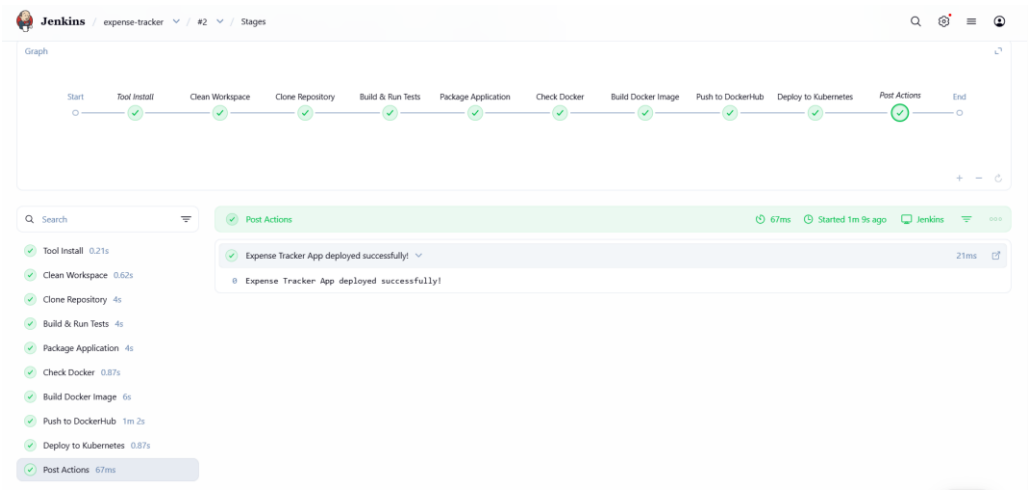
```
}

environment {
    DOCKER_IMAGE = "sandhiya1701/expense-tracker"
    IMAGE_TAG = "${BUILD_NUMBER}"
}

stages {

    stage('Clean Workspace') {
        steps {
            deleteDir()
        }
    }

    stage('Clone Repository') {
        steps {
            git branch: 'main',
            url: 'https://github.com/SandhiyaUmapathi/expense-tracker.git'
        }
    }

    stage('Build & Run Tests') {
        steps {
            bat 'mvn clean test'
        }
        post {
            always {
                junit 'target/surefire-reports/*.xml'
```

```
        }
    }
}

stage('Package Application') {
    steps {
        bat 'mvn clean package -DskipTests'
    }
}

stage('Check Docker') {
    steps {
        bat 'docker info'
    }
}

stage('Build Docker Image') {
    steps {
        bat '''
        docker build -t %DOCKER_IMAGE%:latest .
        docker build -t %DOCKER_IMAGE%:%IMAGE_TAG% .
        '''
    }
}

stage('Push to DockerHub') {
    steps {
        withCredentials([usernamePassword(
            credentialsId: 'dockerhub-creds',
```

```
            usernameVariable: 'DOCKER_USER',
            passwordVariable: 'DOCKER_PASS'
        )]) {
            bat '''
            docker logout

            echo Logging in as %DOCKER_USER%

            docker login -u %DOCKER_USER% -p %DOCKER_PASS%

            IF %ERRORLEVEL% NEQ 0 (
                echo Docker login failed
                exit /b 1
            )

            docker push %DOCKER_IMAGE%:latest
            docker push %DOCKER_IMAGE%:%IMAGE_TAG%
            '''
        }
    }
}

stage('Deploy to Kubernetes') {
    steps {
        bat '''
        echo Setting KUBECONFIG...
        set KUBECONFIG=C:\\Users\\sandh\\.kube\\config

        echo Checking cluster...
```

```
            kubectl get nodes


            echo Deploying application...
            kubectl apply -f deployment.yaml --validate=false
            '''
        }
    }
}


post {
    success {
        echo 'Expense Tracker App deployed successfully!'
    }
    failure {
        echo 'Pipeline failed. Check logs.'
    }
}
}
```

Pipeline Overview



Console Output

Jenkins / expense-tracker / #2

```
Checking cluster...

C:\ProgramData\Jenkins\.jenkins\workspace\expense-tracker>kubectl get nodes
NAME                  STATUS   ROLES           AGE   VERSION
desktop-control-plane   Ready    control-plane   9d    v1.34.3

C:\ProgramData\Jenkins\.jenkins\workspace\expense-tracker>echo Deploying application...
Deploying application...

C:\ProgramData\Jenkins\.jenkins\workspace\expense-tracker>kubectl apply -f deployment.yaml --validate=false
deployment.apps/expense-app created
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Expense Tracker App deployed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Docker Image

docker desktop PERSONAL

Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit BETA
Extensions

Images Give feedback

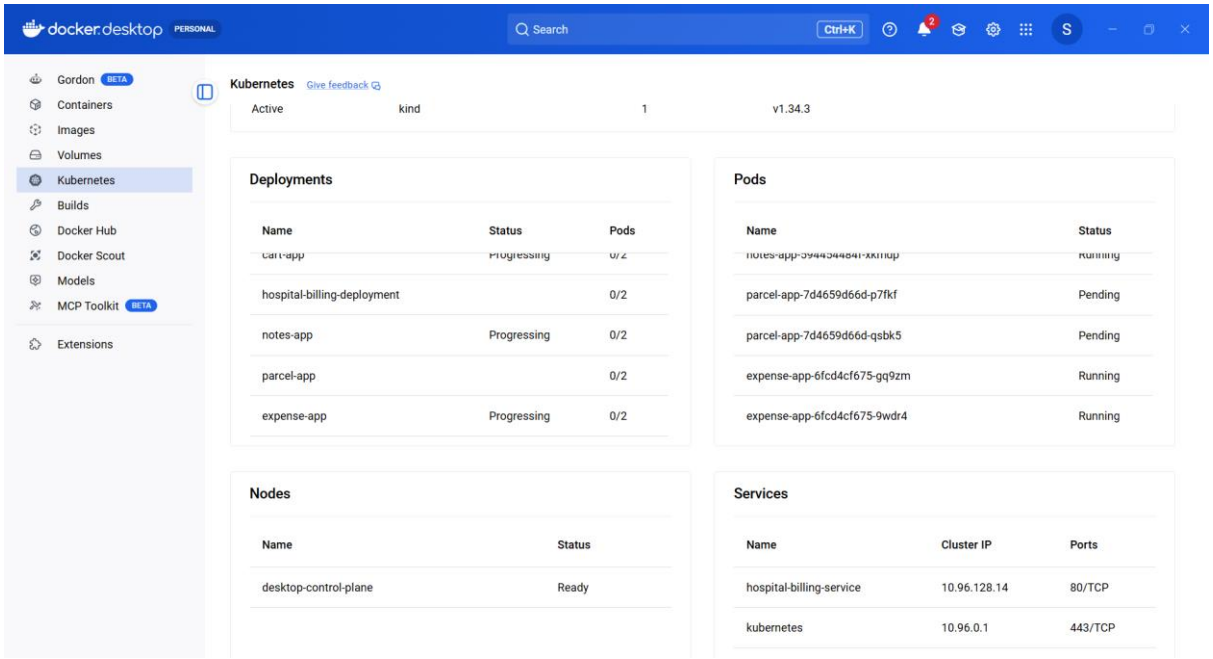
Local My Hub

0 Bytes / 3.39 GB in use 34 images

Last refresh: 3 hours ago

Name	Tag	Image ID	Created	Size
sandhiya1701/java-test-app	8	3ca9a39e83d8	10 days ago	510.06 MB
sandhiya1701/java-test-app	7	f6cef4194970	10 days ago	510.06 MB
sandhiya1701/java-test-app	6	b1af70dae2c6	10 days ago	510.06 MB
sandhiya1701/java-test-app	5	1c122c1d7a03	10 days ago	510.06 MB
sandhiya1701/java-app	latest	632d8f9fb2fb	14 days ago	635.94 MB
kindest/node	v1.34.3	08497ee19eac	4 months ago	1.35 GB
docker/desktop-cloud-provider-kinc	v0.5.0	4ad59ce20658	4 months ago	595.06 MB
envoyproxy/envoy	v1.36.4	08acc53db6f4	4 months ago	247.74 MB
docker/desktop-containerd-registry	v0.0.3	179a9ea3ae47	4 months ago	50.32 MB
sandhiya1701/expense-tracker	1	8a211fe8b51d	9 minutes ago	510.06 MB
sandhiya1701/expense-tracker	latest	7de17ef47dec	4 minutes ago	510.06 MB
sandhiya1701/expense-tracker	2	106f8a4fc20e	4 minutes ago	510.06 MB

Kubernetes



kubectl get svc

```
C:\Users\sandh> kubectl get svc
NAME                      TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
hospital-billing-service  NodePort    10.96.128.14   <none>        80:30007/TCP   7d3h
kubernetes                ClusterIP   10.96.0.1      <none>        443/TCP        9d
```